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Studio Report 2007

Sonic Communications Research Group (SCRG)

<www.scrg.org.au>

About

The Sonic Communications Research Group (SCRG) is an interdisciplinary group of researchers and educators working with sound as a medium for creative communication.

The group has a practice-led approach that seeks to crossover and integrate theory from media studies, design, information science, psychology and the creative arts.

Our special expertise includes sonification, sonic-writing, haptic-audio, auditory interfaces, generative sound, music imagery, music cognition, auditory perception, and sound art, theory and practice.

SCRG is a participant in the ARC Research Network on Human Communication Sciences. <www.hcsnet.edu.au>

People

Faculty
A/Prof. Stephen Barrass (Co-Director, UC)
Prof. Roger Dean (Co-Director, UWS)
Prof Hazel Smith (UWS)
Dr. Mitchell Whitelaw (UC)
Dr. Sam Hinton (UC)

Postdoctoral Fellows
Dr. Freya Bailes (UC)

Postgraduate Students
David Worrall
Dempsey Chang
Robert Bell

Research Assistants
Michael Bylstra

Alumni
Michael Bylstra (hons 2006)
Somaya Langley (hons 2006)

Projects

Sonification of the Capital Markets Worrall/Dean/Barrass/Whitelaw
A pilot projectwith the Capital Markets Collaborative Research Centre (CMCRC).

SoniPy – Worrall/Bylstra/Barrrass/Dean
An open source environment for data sonification research and auditory displays.
sonipy.sourceforge.net

Percussive Instrument Timbre Classification Hierarchy Bell/Barrass
A system for analysing and classifying the timbres of percussion instruments.
percussionresearch.uniblogs.org

Realtime Computational Generation of Large Scale Musical Structure, ARC Discovery Grant, 2004-6. Dean/Smith/Bailes
The project aims to develop efficient techniques for the generation and control of large-scale musical structure in real-time, and to use these to assess the degree to which the resultant structures are cognitively accessible to experienced musicians.

Cloud Musak - Bylstra/Barrass
An ambient sonification of clouds and sky taken from webcam and internet weather stations.
cloudmusak.blogspot.com

The Suspect Backpack - Langley/Whitelaw
A wearable mobile sonic media art experience, providing first-person interaction and engagement in public and private spaces. This [prototype] work utilises proximity sensors, a laptop, a pair of speakers and a set of headphones to navigate two spoken-word sound environments.
suspectbackpack.blogspot.com

Audio-Visual Augmented Reality (AVIARy) – Barrass/Morse
An Augmented Reality system for moulding musical and visual gestalts from algorithmic materials layered in space and time.
http://www.petermorse.com.au/old/PMcomhtml/petermorse.html

Creative Works

These are just a sampling of performances, installations and other other creative audio works by members of SCRG in recent times. There are significant works that I haven't yet collated in time for this studio report. I will have a more complete collection to show at ACMA.

Langley S. (2007) The Suspect Backpack, Dorkbot, Melbourne, May 2007
criticalsenses.blogspot.com

Dean R. (2006) Sonic Intermedia Performance, Cultural Studies Association of Australasia Annual Conference, December 2006.
www.unaustralia.com/speakers.php

Barrass, S. (2006) OpShop, Experimenta Under the Radar, Institute for Contemporary Art, London, December 2006
www.ica.org.uk/Experimenta%20Under%20the%20Radar+12289.twl

Barrass, S., Bylstra M. and Kuzmanovic, M. (2006) Fruit Turntable Custard, S-crepe Happenings, Canberra, January 2006
www.criticalsenses.com/screpe

Whitelaw, M. and Gaffney N. (2006) Boom, S-crepe Happenings, Canberra, January 2006
www.criticalsenses.com/screpe

Barrass, S. (2005) Augmented Reality Crazy Frog phone-tone remix, Bimbimbie Sound and Intermedia Performance, University of Canberra, October 2005

Dean, R. and Smith, H. (2005) Time the Magician, Bimbimbie Sound and Intermedia Performance, University of Canberra, October 2005

Whitelaw, M. (2005) Arcadia Sonic Mosaic, Sound and Intermedia Performance, University of Canberra, October 2005

Barrass, S., Riddell, A., Fitton, A., Barrass T., Morse, P., and Onaclov (2005) Edible Audience, Liquid Architecture 6, National Gallery of Australia, Canberra, July 2005.
www.nga.gov.au/LiquidArchitecture

Publications

Bailes, Freya (2007). A response to Andrea Halpern's commentary. *Empirical Musicology Review*. 2(2), 62-64.

Barrass, Stephen and Whitelaw, Mitchell and Bailes, Freya (2007). Listening to the Mind Listening: An Analysis of Sonification Reviews, Designs and Correpondences. *Leonardo Music Journal*. 16, 13-19.

Bailes, Freya (2007). Timbre as an elusive component of imagery for music.. *Empirical Musicology Review*. 2(1), 21-34.

Barrass, Stephen, Barrass, Tim (2006). Musical creativity in collaborative virtual environments. *Virtual Reality*. 10, 149--157.

Barrass, Stephen, Whitelaw, Mitchell, Potard, Guillaume (2006). Listening to the mind listening. *Media International Australia incorporating Culture and Policy*. 2006, 60-67.

Dean, Roger, Whitelaw, Mitchell, Smith, Hazel, Worrall, David (2006). The mirage of real-time algorithmic synaesthesia: Some compositional mechanisms and research agendas in computer music and sonification. *Contemporary Music Review*. 25, 311--326.

Barrass, Stephen (2006). Haptic-Audio Narrative: From Physical Simulation to Imaginative Stimulation. Springer *Lecture Notes in Computer Science*. 4129, 157-165.

Barrass, Stephen (2005). A perceptual framework for the auditory display of scientific data. *ACM Trans. Appl. Percept.*. 2, 389--402.

Bailes, Freya, Dean, Roger T. (2005). Structural Judgements in the Perception of Computer-Generated Music. *2nd International Conference of the Asia Pacific Society for the Cognitive Science of Music* (APSCOM). 155-160.

Whitelaw, Mitchell (2005). System Stories and Model Worlds: A Critical Approach to Generative Art. *Readme 100: Temporary Software Art Factory*. 135--154.

Whitelaw, Mitchell (2004). Hearing Pure Data: Aesthetics and Ideals of Data-Sound. *Unsorted: Thoughts on the Information Arts: An A to Z for Sonic Acts X*. 45--54.

Whitelaw, Mitchell (2004). *Metacreation: Art and Artificial Life*, MIT Press.

Nesbitt, K. V., Barrass, S. (2004). Finding trading patterns in stock market data. *Computer Graphics and Applications*, IEEE. 24, 45--55. Abstract

Whitelaw, Mitchell (2003). Sound Particles and Microsonic Materialism. *Contemporary Music Review*. 22, 93--100.

Barrass, Stephen (1998). Auditory Information Design. Ph.D. Thesis. Australian National University